Exploring Faculty Perceptions and Concerns Regarding Artificial Intelligence Chatbots in Nursing Education: Potential Benefits and Limitations

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Abstract

This study aims to examine faculty perceptions of artificial intelligence (AI) chatbots in nursing education, focusing on their usage patterns, perceived benefits, and limitations. A cross-sectional study. The study surveyed nursing faculty from Jordan and the United States using a self-reported questionnaire. Data were analyzed using descriptive statistics and Multivariate Analysis of Covariance to assess variations in perceptions based on AI chatbot usage frequency and faculty characteristics. Among 474 faculty members, 82.5% were familiar with at least one AI chatbot. Faculty generally acknowledged the benefits of AI chatbots, including enhanced teaching experiences, improved student engagement, support for independent learning, and quick access to medical knowledge. However, concerns about misinformation, reduced faculty-student interaction, and inadequacies in addressing complex clinical scenarios were prevalent. Legal and ethical issues, particularly the risk of misuse of AI-generated information, were also highlighted. Frequent AI chatbot users demonstrated significantly greater awareness of both the

advantages and limitations of AI chatbots compared to infrequent users. Frequent users demonstrated greater awareness of both the advantages and challenges of AI chatbots, highlighting the role of hands-on experience in shaping adoption. However, faculty adoption is primarily driven by perceived benefits rather than limitations, emphasizing the need to showcase practical advantages while addressing concerns. To enhance faculty adoption of AI chatbots in nursing education, institutions should focus on demonstrating their practical benefits while addressing concerns through targeted training and ethical guidelines. Providing hands-on experience and structured exposure can increase faculty confidence, reinforcing both the usefulness of AI chatbots and strategies to mitigate their limitations. Future research may examine the effectiveness of hands-on training programs in shaping faculty perceptions and adoption behaviors, providing valuable insights for enhancing AI integration in nursing education.